Georgia Tech Bus Analytics Report

Data from 2026-03-16 14:50:19 to 2026-03-16 17:46:22 — 5,176 polls across 18 buses

1. Route Overview

Route	Buses	# Buses	Total Polls	First Poll	Last Poll
Red Red	2201, 2202, 2206	3	882	2026-03-16 14:50:19	2026-03-16 17:46:22
Blue Blue	2203, 2208, 2214	3	882	2026-03-16 14:50:19	2026-03-16 17:46:22
Gold Gold	2301, 2302, 2304, 2307	4	1176	2026-03-16 14:50:19	2026-03-16 17:46:22

2. Stop & Break Analysis

A 'stop' is a consecutive run of polls with speed = 0. Duration = count × 35s poll interval.

Red Red Route

Bus	# Stops	Total Break Time	Avg Break	Longest Break
2201	56	94m 30s	1m 41s	18m 40s
2202	56	99m 10s	1m 46s	29m 10s
2206	60	98m	1m 38s	35m 35s

▶ **2201** — 56 stop(s)
▶ **2202** — 56 stop(s)
▶ **2206** — 60 stop(s)

Blue Blue Route

Bus	# Stops	Total Break Time	Avg Break	Longest Break
2203	58	89m 15s	1m 32s	5m 50s
2208	54	73m 30s	1m 21s	11m 5s
2214	64	87m 30s	1m 22s	7m 35s

- **2203** — 58 stop(s)
- **2208** — 54 stop(s)
- **2214** — 64 stop(s)

Gold **Gold Route**

Bus	# Stops	Total Break Time	Avg Break	Longest Break
2301	81	88m 5s	1m 5s	6m 25s
2302	58	83m 25s	1m 26s	9m 55s
2304	60	85m 10s	1m 25s	10m 30s
2307	50	69m 25s	1m 23s	5m 50s

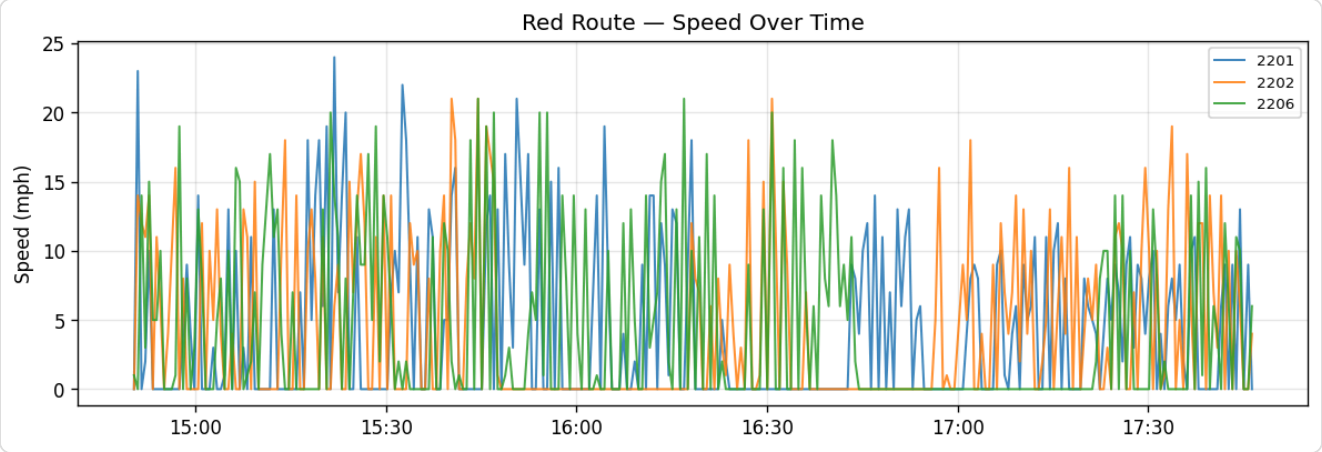
- **2301** — 81 stop(s)
- **2302** — 58 stop(s)
- **2304** — 60 stop(s)
- **2307** — 50 stop(s)

3. Speed Profiles

Average speed excludes polls where bus was stopped (speed = 0).

Red **Red Route**

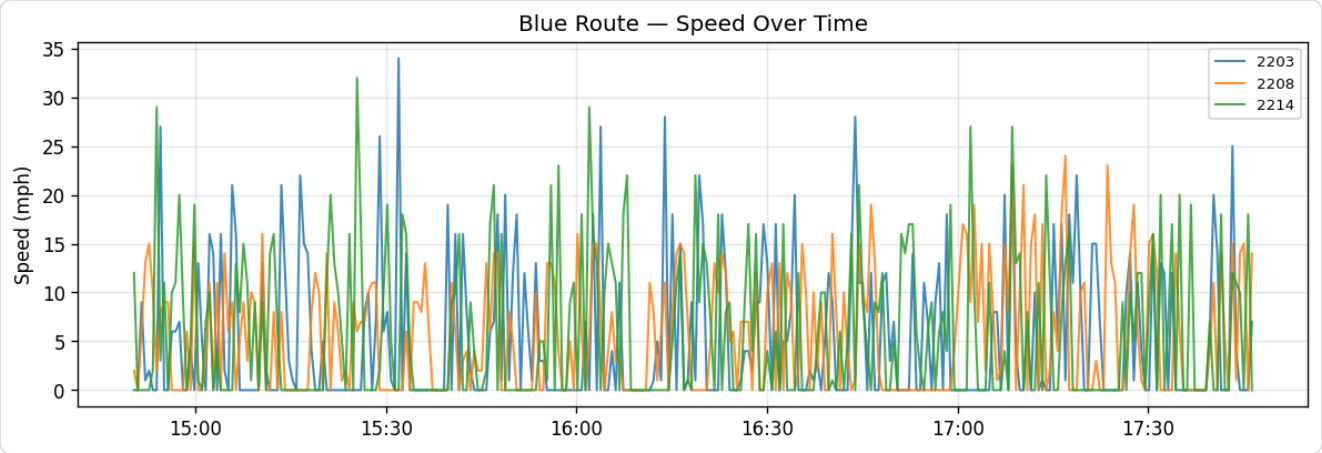
Bus	Avg Speed	Max Speed	Min Speed	% Moving	Total Polls
2201	9.5	24.0	1.0	45%	294
2202	9.9	21.0	1.0	42%	294
2206	9.8	21.0	1.0	43%	294



Blue

Blue Route

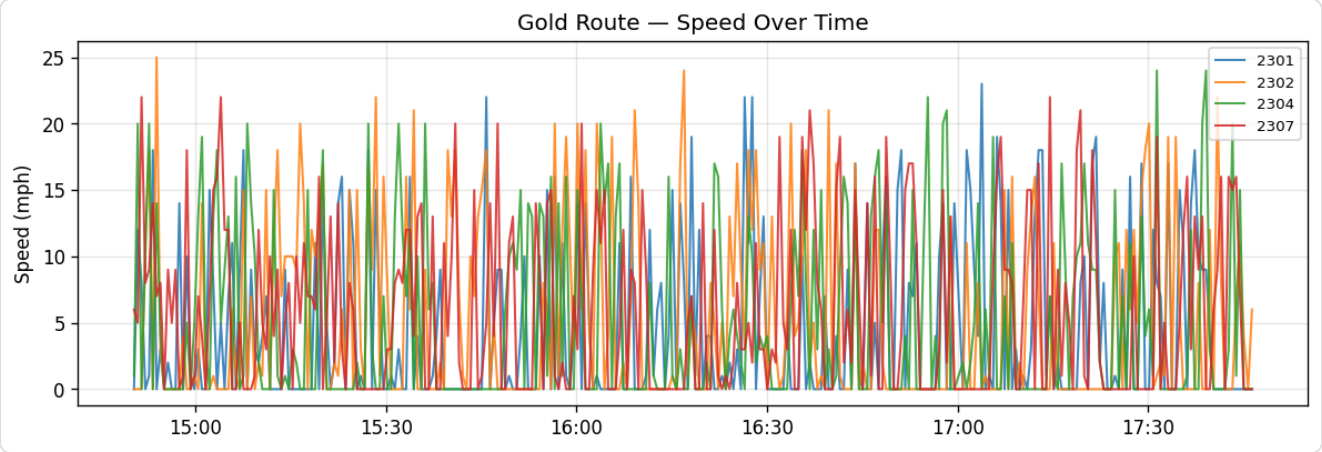
Bus	Avg Speed	Max Speed	Min Speed	% Moving	Total Polls
2203	10.6	34.0	1.0	48%	294
2208	9.5	24.0	1.0	57%	294
2214	11.7	32.0	1.0	49%	294



Gold

Gold Route

Bus	Avg Speed	Max Speed	Min Speed	% Moving	Total Polls
2301	9.3	23.0	1.0	49%	294
2302	10.1	25.0	1.0	51%	294
2304	10.0	24.0	1.0	50%	294
2307	9.6	22.0	1.0	60%	294



4. Inter-Bus Distance & Propagation Delay

Pairwise haversine distance between buses on the same route at each poll. The **Heading** field classifies each measurement:

- **Same direction** (≤60° apart): buses traveling together — close distance = bunching
- **Opposite direction** (≥120° apart): buses passing each other — close distance is expected
- **Angled** (60°–120°): turning / diverging
- **Unknown**: one or both buses stopped (heading = 0)

Red **Red Route**

Bus Pair	Mean Distance	Min (bunching)	Max	Samples
2201 ↔ 2202	816 m	0 m	1.69 km	294
2201 ↔ 2206	868 m	6 m	1.74 km	294
2202 ↔ 2206	887 m	0 m	1.72 km	294

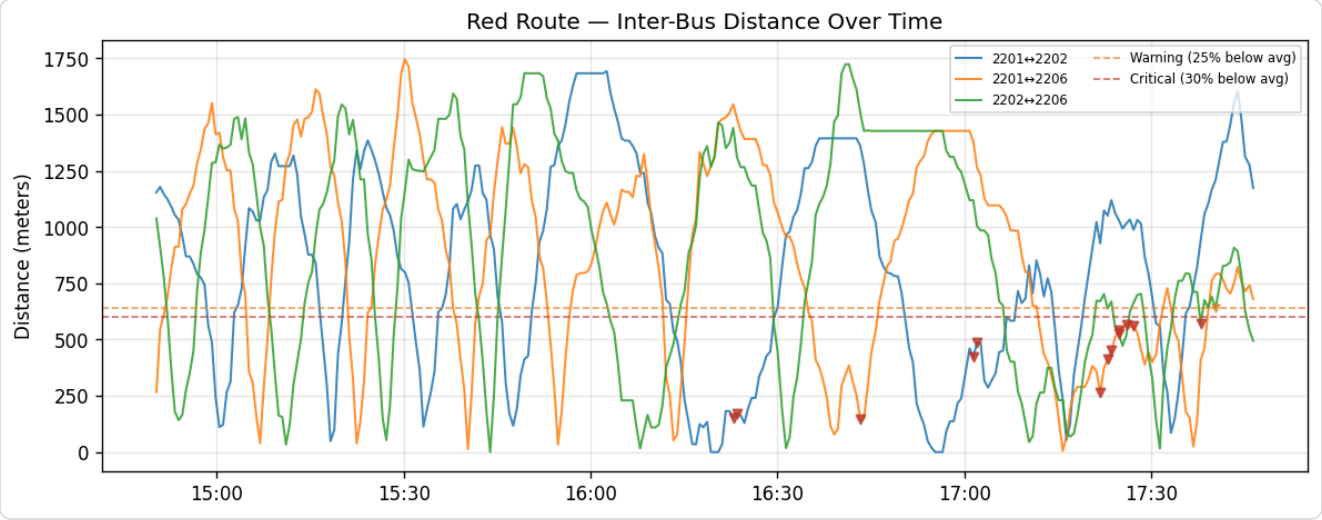
857 m
Avg Spacing

0 m
Min Spacing (bunching)

1.74 km
Max Spacing

1
Spacing Warnings (25%↓)

13
Spacing Critical (30%↓)



Spacing Alerts

Timestamps where a bus pair's distance fell 25%+ below that pair's average. Orange ▼ = warning (25-30% below), Red ▼ = critical (30%+ below).

Time	Pair	Distance	Pair Avg	% Below	Direction	Severity
2026-03-16 16:22:55	2201 ↔ 2202	152 m	816 m	81%	Same direction	**CRITICAL**
2026-03-16 16:23:30	2201 ↔ 2202	172 m	816 m	79%	Same direction	**CRITICAL**
2026-03-16 16:43:18	2201 ↔ 2206	149 m	868 m	83%	Same direction	**CRITICAL**
2026-03-16 17:01:26	2201 ↔ 2202	422 m	816 m	48%	Same direction	**CRITICAL**
2026-03-16 17:02:02	2201 ↔ 2202	487 m	816 m	40%	Same direction	**CRITICAL**
2026-03-16 17:21:49	2201 ↔ 2206	263 m	868 m	70%	Same direction	**CRITICAL**
2026-03-16 17:23:00	2201 ↔ 2206	412 m	868 m	53%	Same direction	**CRITICAL**
2026-03-16 17:23:36	2201 ↔ 2206	451 m	868 m	48%	Same direction	**CRITICAL**
2026-03-16 17:24:48	2201 ↔ 2206	542 m	868 m	37%	Same direction	**CRITICAL**

Time	Pair	Distance	Pair Avg	% Below	Direction	Severity
2026-03-16 17:24:48	2202 ↔ 2206	525 m	887 m	41%	Same direction	CRITICAL
2026-03-16 17:25:59	2201 ↔ 2206	567 m	868 m	35%	Same direction	CRITICAL
2026-03-16 17:27:11	2201 ↔ 2206	562 m	868 m	35%	Same direction	CRITICAL
2026-03-16 17:37:57	2202 ↔ 2206	570 m	887 m	36%	Same direction	CRITICAL
2026-03-16 17:40:19	2202 ↔ 2206	638 m	887 m	28%	Same direction	WARNING

Heading Analysis

Direction	Samples	% of Total	Avg Dist	Min Dist	Max Dist
Same direction	34	4%	783 m	149 m	1.46 km
Opposite direction	61	7%	526 m	12 m	1.71 km
Angled	57	6%	924 m	89 m	1.55 km
Unknown (stopped)	730	83%	883 m	0 m	1.74 km

34
Same-Dir Samples

783 m
Avg Same-Dir Spacing

3
Bunching Events (≤200m, same dir)

> **3** times two buses were within 200m *and* heading the same direction — true bunching.

Blue **Blue Route**

Bus Pair	Mean Distance	Min (bunching)	Max	Samples
2203 ↔ 2208	865 m	8 m	1.69 km	294
2203 ↔ 2214	584 m	0 m	1.46 km	294

Bus Pair	Mean Distance	Min (bunching)	Max	Samples
2208 ↔ 2214	726 m	0 m	1.68 km	294

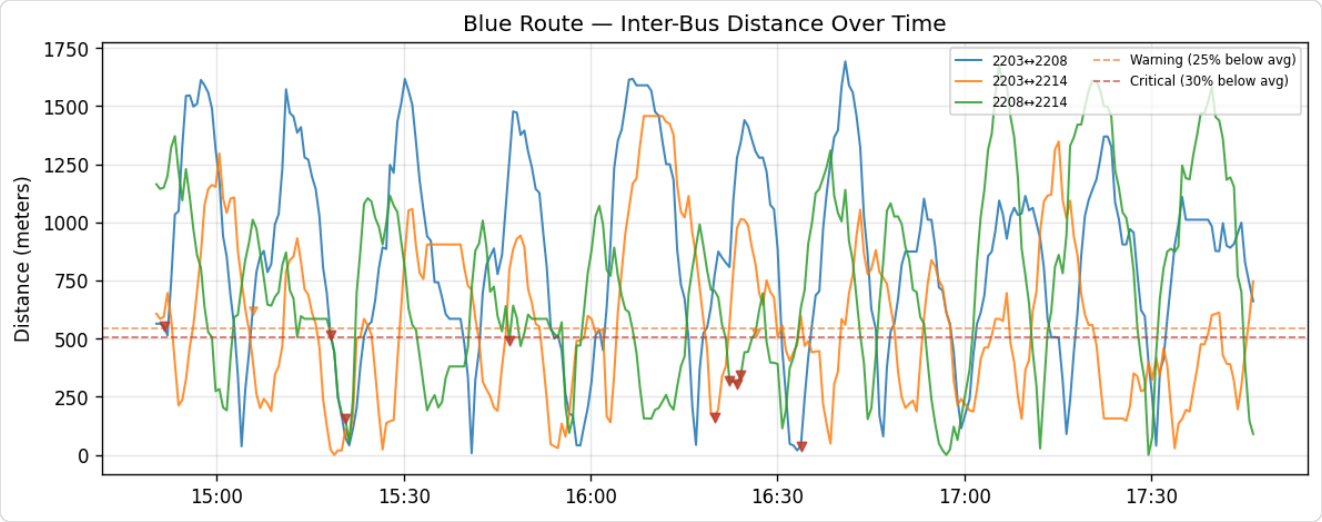
725 m
Avg Spacing

0 m
Min Spacing (bunching)

1.69 km
Max Spacing

2
Spacing Warnings (25%↓)

9
Spacing Critical (30%↓)



Spacing Alerts

Timestamps where a bus pair's distance fell 25%+ below that pair's average. Orange ▼ = warning (25-30% below), Red ▼ = critical (30%+ below).

Time	Pair	Distance	Pair Avg	% Below	Direction	Severity
2026-03-16 14:51:30	2203 ↔ 2208	556 m	865 m	36%	Same direction	CRITICAL
2026-03-16 15:05:46	2203 ↔ 2208	622 m	865 m	28%	Same direction	WARNING
2026-03-16 15:18:16	2203 ↔ 2208	513 m	865 m	41%	Same direction	CRITICAL
2026-03-16 15:20:39	2208 ↔ 2214	155 m	726 m	79%	Same direction	CRITICAL

Time	Pair	Distance	Pair Avg	% Below	Direction	Severity
2026-03-16 15:46:59	2208 ↔ 2214	491 m	726 m	32%	Same direction	CRITICAL
2026-03-16 16:19:55	2203 ↔ 2214	159 m	584 m	73%	Same direction	CRITICAL
2026-03-16 16:22:19	2208 ↔ 2214	317 m	726 m	56%	Same direction	CRITICAL
2026-03-16 16:23:30	2208 ↔ 2214	304 m	726 m	58%	Same direction	CRITICAL
2026-03-16 16:24:06	2208 ↔ 2214	341 m	726 m	53%	Same direction	CRITICAL
2026-03-16 16:26:29	2208 ↔ 2214	526 m	726 m	28%	Same direction	WARNING
2026-03-16 16:33:46	2203 ↔ 2208	39 m	865 m	95%	Same direction	CRITICAL

Heading Analysis

Direction	Samples	% of Total	Avg Dist	Min Dist	Max Dist
Same direction	59	7%	832 m	39 m	1.62 km
Opposite direction	91	10%	649 m	8 m	1.58 km
Angled	78	9%	740 m	20 m	1.58 km
Unknown (stopped)	654	74%	724 m	0 m	1.69 km

59
Same-Dir Samples

832 m
Avg Same-Dir Spacing

3
Bunching Events (≤200m, same dir)

> **3** times two buses were within 200m *and* heading the same direction — true bunching.

Gold **Gold Route**

Bus Pair	Mean Distance	Min (bunching)	Max	Samples
2301 ↔ 2302	865 m	9 m	1.79 km	294
2301 ↔ 2304	974 m	11 m	1.77 km	294
2301 ↔ 2307	732 m	20 m	1.63 km	294
2302 ↔ 2304	614 m	0 m	1.52 km	294
2302 ↔ 2307	939 m	0 m	1.73 km	294
2304 ↔ 2307	821 m	17 m	1.65 km	294

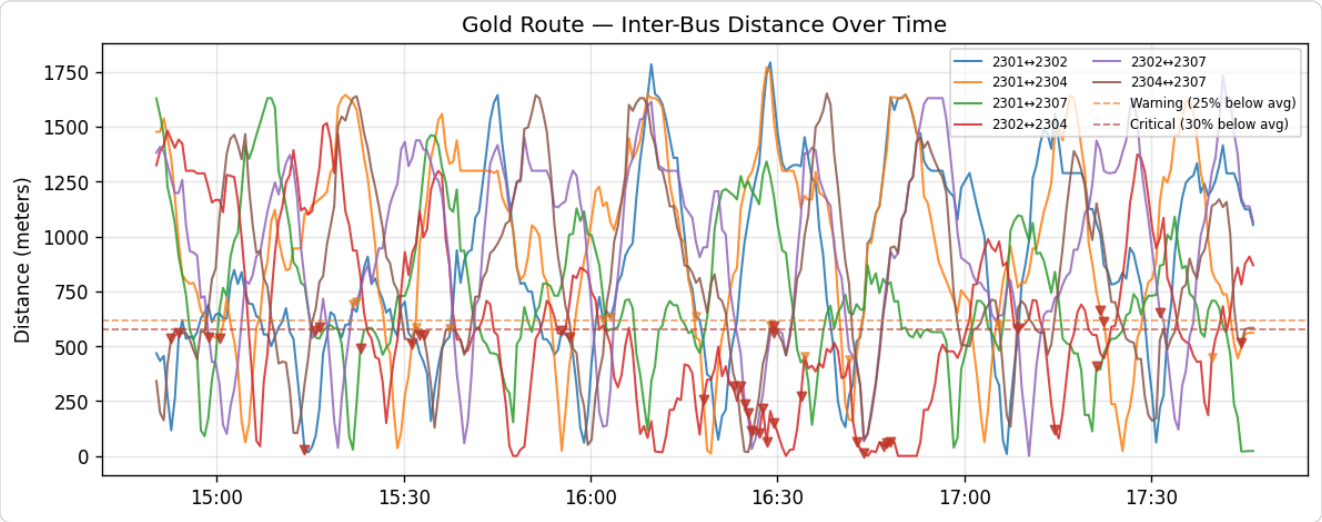
824 m
Avg Spacing

0 m
Min Spacing (bunching)

1.79 km
Max Spacing

12
Spacing Warnings (25%↓)

38
Spacing Critical (30%↓)



Spacing Alerts

Timestamps where a bus pair's distance fell 25%+ below that pair's average. Orange ▼ = warning (25-30% below), Red ▼ = critical (30%+ below).

Time	Pair	Distance	Pair Avg	% Below	Direction	Severity
2026-03-16 14:52:41	2304 ↔ 2307	539 m	821 m	34%	Same direction	CRITICAL
2026-03-16 14:53:52	2304 ↔ 2307	562 m	821 m	31%	Same direction	CRITICAL
2026-03-16 14:58:38	2301 ↔ 2304	540 m	974 m	45%	Same direction	CRITICAL
2026-03-16 15:00:25	2301 ↔ 2304	537 m	974 m	45%	Same direction	CRITICAL
2026-03-16 15:14:06	2301 ↔ 2302	26 m	865 m	97%	Same direction	CRITICAL
2026-03-16 15:15:53	2302 ↔ 2307	569 m	939 m	39%	Same direction	CRITICAL
2026-03-16 15:16:28	2302 ↔ 2307	586 m	939 m	38%	Same direction	CRITICAL
2026-03-16 15:21:52	2302 ↔ 2307	695 m	939 m	26%	Same direction	WARNING
2026-03-16 15:22:27	2302 ↔ 2307	704 m	939 m	25%	Same direction	WARNING
2026-03-16 15:23:03	2301 ↔ 2307	488 m	732 m	33%	Same direction	CRITICAL
2026-03-16 15:31:23	2304 ↔ 2307	513 m	821 m	37%	Same direction	CRITICAL
2026-03-16 15:31:59	2304 ↔ 2307	589 m	821 m	28%	Same direction	WARNING
2026-03-16 15:32:35	2304 ↔ 2307	553 m	821 m	33%	Same direction	CRITICAL
2026-03-16 15:33:10	2304 ↔ 2307	552 m	821 m	33%	Same direction	CRITICAL
2026-03-16 15:37:21	2304 ↔ 2307	581 m	821 m	29%	Same direction	WARNING
2026-03-16 15:55:22	2301 ↔ 2302	570 m	865 m	34%	Same direction	CRITICAL

Time	Pair	Distance	Pair Avg	% Below	Direction	Severity
2026-03-16 15:56:34	2301 ↔ 2302	541 m	865 m	37%	Same direction	CRITICAL
2026-03-16 16:02:00	2301 ↔ 2302	622 m	865 m	28%	Same direction	WARNING
2026-03-16 16:03:12	2301 ↔ 2302	632 m	865 m	27%	Same direction	WARNING
2026-03-16 16:16:56	2301 ↔ 2302	637 m	865 m	26%	Same direction	WARNING
2026-03-16 16:18:07	2302 ↔ 2304	255 m	614 m	58%	Same direction	CRITICAL
2026-03-16 16:22:55	2302 ↔ 2304	316 m	614 m	48%	Same direction	CRITICAL
2026-03-16 16:24:06	2302 ↔ 2304	317 m	614 m	48%	Same direction	CRITICAL
2026-03-16 16:24:42	2302 ↔ 2304	237 m	614 m	61%	Same direction	CRITICAL
2026-03-16 16:25:18	2302 ↔ 2304	195 m	614 m	68%	Same direction	CRITICAL
2026-03-16 16:25:53	2302 ↔ 2304	117 m	614 m	81%	Same direction	CRITICAL
2026-03-16 16:27:04	2302 ↔ 2304	105 m	614 m	83%	Same direction	CRITICAL
2026-03-16 16:27:40	2302 ↔ 2304	217 m	614 m	65%	Same direction	CRITICAL
2026-03-16 16:28:16	2302 ↔ 2304	65 m	614 m	89%	Same direction	CRITICAL
2026-03-16 16:28:51	2304 ↔ 2307	596 m	821 m	27%	Same direction	WARNING
2026-03-16 16:29:27	2302 ↔ 2304	152 m	614 m	75%	Same direction	CRITICAL
2026-03-16 16:29:27	2302 ↔ 2307	594 m	939 m	37%	Same direction	CRITICAL

Time	Pair	Distance	Pair Avg	% Below	Direction	Severity
2026-03-16 16:29:27	2304 ↔ 2307	564 m	821 m	31%	Same direction	CRITICAL
2026-03-16 16:33:46	2302 ↔ 2304	272 m	614 m	56%	Same direction	CRITICAL
2026-03-16 16:34:21	2302 ↔ 2304	456 m	614 m	26%	Same direction	WARNING
2026-03-16 16:41:29	2302 ↔ 2304	437 m	614 m	29%	Same direction	WARNING
2026-03-16 16:42:41	2302 ↔ 2304	67 m	614 m	89%	Same direction	CRITICAL
2026-03-16 16:43:53	2302 ↔ 2304	15 m	614 m	97%	Same direction	CRITICAL
2026-03-16 16:46:59	2302 ↔ 2304	43 m	614 m	93%	Same direction	CRITICAL
2026-03-16 16:47:35	2302 ↔ 2304	58 m	614 m	90%	Same direction	CRITICAL
2026-03-16 16:48:10	2302 ↔ 2304	64 m	614 m	90%	Same direction	CRITICAL
2026-03-16 17:05:36	2304 ↔ 2307	599 m	821 m	27%	Same direction	WARNING
2026-03-16 17:08:36	2302 ↔ 2307	581 m	939 m	38%	Same direction	CRITICAL
2026-03-16 17:14:33	2302 ↔ 2304	122 m	614 m	80%	Same direction	CRITICAL
2026-03-16 17:21:13	2301 ↔ 2307	407 m	732 m	44%	Same direction	CRITICAL
2026-03-16 17:21:49	2301 ↔ 2304	665 m	974 m	32%	Same direction	CRITICAL
2026-03-16 17:22:25	2301 ↔ 2304	611 m	974 m	37%	Same direction	CRITICAL
2026-03-16 17:31:23	2302 ↔ 2307	654 m	939 m	30%	Same direction	CRITICAL

Time	Pair	Distance	Pair Avg	% Below	Direction	Severity
2026-03-16 17:39:44	2302 ↔ 2304	448 m	614 m	27%	Same direction	WARNING
2026-03-16 17:44:29	2304 ↔ 2307	514 m	821 m	37%	Same direction	CRITICAL

Heading Analysis

Direction	Samples	% of Total	Avg Dist	Min Dist	Max Dist
Same direction	134	8%	825 m	15 m	1.79 km
Opposite direction	194	11%	709 m	17 m	1.64 km
Angled	165	9%	860 m	66 m	1.65 km
Unknown (stopped)	1271	72%	837 m	0 m	1.78 km

134
Same-Dir Samples

825 m
Avg Same-Dir Spacing

12
Bunching Events (≤200m, same dir)

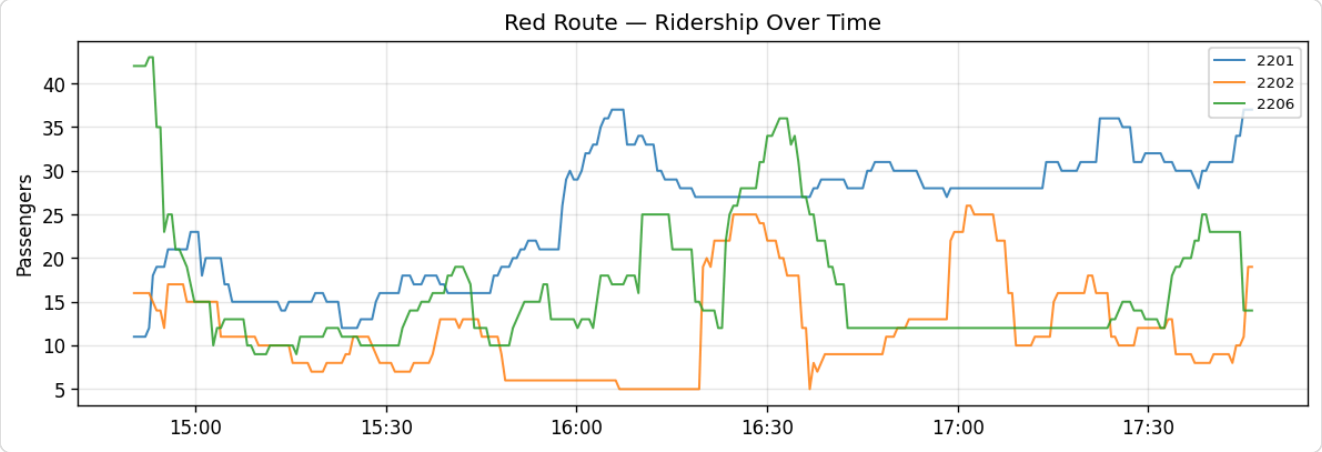
> **12** times two buses were within 200m *and* heading the same direction — true bunching.

5. Capacity & Ridership

Data from `GetVehicleCapacities`: current occupation vs. seat capacity per bus.

Red **Red Route**

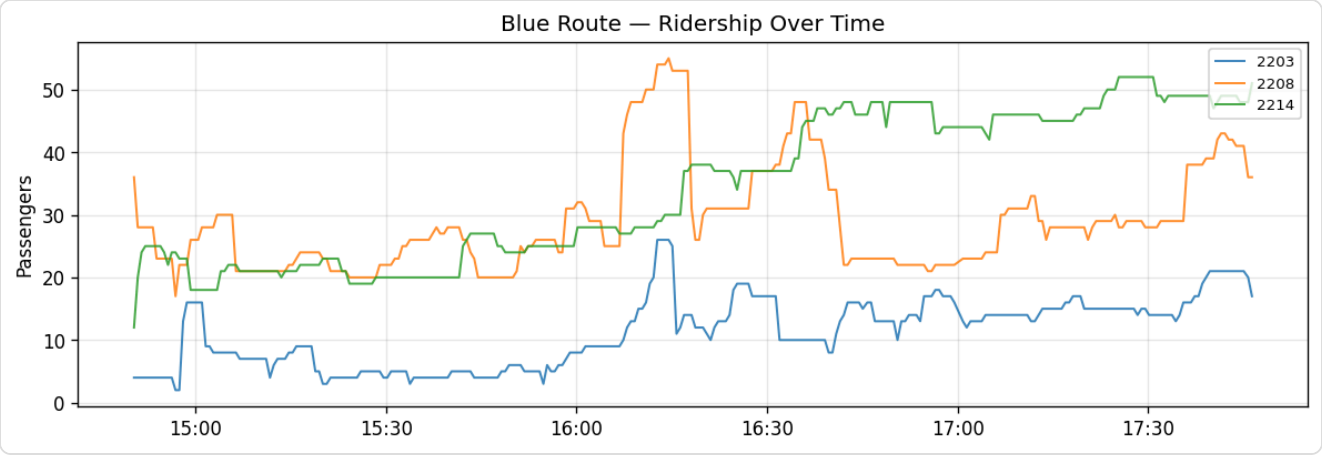
Bus	Capacity	Avg Riders	Peak Riders	Min Riders	Avg Util%	Peak Util%
2201	40	25	37	11	62%	92%
2202	40	12	26	5	30%	65%
2206	40	16	43	9	41%	108%



Blue

Blue Route

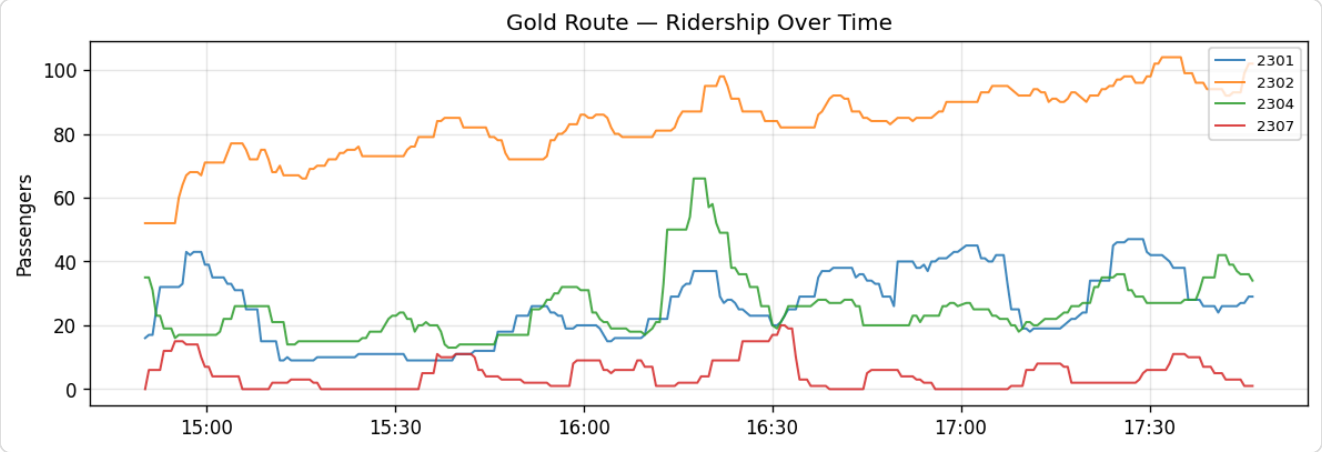
Bus	Capacity	Avg Riders	Peak Riders	Min Riders	Avg Util%	Peak Util%
2203	40	11	26	2	28%	65%
2208	40	29	55	17	73%	138%
2214	40	34	52	12	86%	130%



Gold

Gold Route

Bus	Capacity	Avg Riders	Peak Riders	Min Riders	Avg Util%	Peak Util%
2301	60	26	47	9	43%	78%
2302	45	83	104	52	184%	231%
2304	45	25	66	13	56%	147%
2307	45	5	20	0	10%	44%


Gold Route — Ridership Over Time
Passengers
0
20
40
60
80
100
15:00
15:30
16:00
16:30
17:00
17:30
2301
2302
2304
2307